Ising Model VQE and Adiabatic Quantum Evolution with Noise

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(Dated: October 20, 2026)

This project compares the variational quantum eigensolver (VQE) and digitized adiabatic quantum evolution for ground-state preparation in the transverse-field Ising model. Both algorithms are implemented and simulated classically using IBM's Qiskit, with and without simulated quantum noise. The results show that VQE achieves small ground-state energy errors, while adiabatic evolution achieves higher GHZ fidelity in the ideal case. With noise, however, the fidelity of adiabatic evolution generally decreases as the number of Trotter steps increases. The highest observed GHZ fidelity with noise is $F_{\mathrm{GHZ}} = 0.951$, achieved with $T = 10$ and $M = 15$.

INTRODUCTION

The Ising model describes interacting spins on a lattice, commonly with nearest-neighbor interactions. Introducing a transverse magnetic field gives rise to the transverse-field Ising model, which provides a simple system for studying quantum many-body physics.

Here, the model is used to compare two approaches to quantum ground-state preparation: the variational quantum eigensolver (VQE) and adiabatic quantum evolution.

MODEL AND METHODS

The transverse-field Ising Hamiltonian is given by

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_i Z_i - g \sum_i X_i, \tag{1}$$

where X_i and Z_i are Pauli operators acting on spin site i, J is the interaction strength, and h and g are the longitudinal and transverse field strengths, respectively.

The parameterized circuit used for state preparation starts from n qubits initialized in $|0\rangle^{\otimes n}$. A unitary $U(\vec{\theta})$, illustrated in Fig. 1, prepares the state

$$|\psi(\vec{\theta})\rangle = U(\vec{\theta})|0\rangle^{\otimes n}. \tag{2}$$

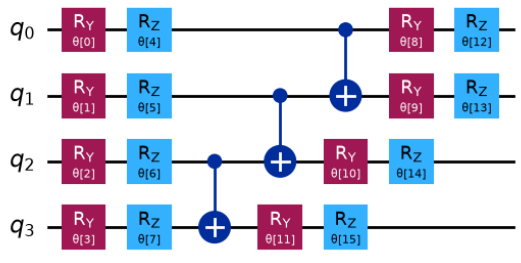


FIG. 1. State preparation circuit for four qubits.

For example, $\vec{\theta} = \vec{0}$ gives $|\psi(\vec{\theta})\rangle = |0\rangle^{\otimes n}$ and $\vec{\theta} = (\pi/2, ..., \pi/2, 0, ..., 0)$, where there are n angles equal $\pi/2$, and $3n$ angles equal 0 ($\vec{\theta}$ has $4n$ parameters), gives $|\psi(\vec{\theta})\rangle = |+\rangle^{\otimes n}$.

Variational quantum eigensolver

VQE uses the variational principle to estimate the ground-state energy E_0:

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle \geq E_0. \tag{3}$$

The circuit in Fig. 1 prepares $|\psi(\boldsymbol{\theta})\rangle$, while a classical optimizer updates $\vec{\theta}$ to minimize the energy expectation value.

Adiabatic quantum evolution

Adiabatic evolution starts in the known ground state of an initial Hamiltonian. Here, we consider

$$H_i = -g_i \sum_i X_i, \tag{4}$$

whose ground state is $|+\rangle^{\otimes N}$ for $g_i > 0$.

The Hamiltonian is gradually changed to the target Hamiltonian H_f according to

$$H(s) = (1-s)H_i + sH_f, \qquad s = t/T, \tag{5}$$

where T is the total evolution time. For sufficiently slow evolution and a large enough spectral gap, the final state approaches the ground state of H_f. Here, H is given by (1).

The evolution is discretized into M time steps of duration $\Delta t = T/M$. At each step, the evolution operator is approximated using a first-order Lie–Trotter decomposition (see Qiskit's LieTrotter). For a Hamiltonian $H = H_1 + H_2$,

$$e^{-i\Delta t(H_1+H_2)} = e^{-i\Delta t H_1} e^{-i\Delta t H_2} + \mathcal{O}(\Delta t^2). \tag{6}$$

The approximation becomes more accurate for smaller time steps, at the cost of increasing the number of quantum gates.

Code

The code is available on GitHub [1], with instructions for installation and execution provided in the README.

The transverse-field Ising Hamiltonian is implemented using Qiskit, supporting both one- and two-dimensional lattices. VQE and digitized adiabatic quantum evolution are simulated classically using Qiskit Aer. For comparison, the ground-state energy is also calculated by exact diagonalization.

Noise

Quantum noise is introduced using Qiskit Aer's noise simulation tools. The implementation supports several noise channels, including depolarizing errors, bit- and phase-flip errors, amplitude and phase damping, and thermal relaxation. These can be applied with user-defined parameters to investigate their influence on the algorithms.

Alternatively, a noise model can be constructed from IBM's 127-qubit `FakeSherbrooke` backend [2], which provides hardware-derived gate errors and relaxation parameters. This allows the algorithms to be investigated under more realistic, although still simulated, noise conditions.

The quantum circuits are transpiled to match the basis gates of the backend. However, the simulations are not intended to reproduce the performance of the physical device quantitatively, since its qubit connectivity and physical qubit layout are not fully accounted for. Further details on the implemented noise channels and their parameters are provided in the README [1].

RESULTS

Comparison of VQE and adiabatic evolution

The ground-state energies obtained using VQE and adiabatic quantum evolution (AQC) are compared with the exact ground-state energy E_0, calculated by numerical diagonalization. The energy error is defined as

$$\Delta E = |E - E_0|. \tag{7}$$

For the two-dimensional four-qubit Ising model with $J = 1$, $h = 0$, and $g = 0.01$, the exact ground-state energy is approximately $E_0 = -4.0001$.

In this regime, where $g \ll J$ and $h = 0$, the ground state is approximately the GHZ state

$$\left|\mathrm{GHZ}_n\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0\right\rangle^{\otimes n} + \left|1\right\rangle^{\otimes n}\right). \tag{8}$$

The fidelity of the final AQC state is therefore also calculated as

$$F_{\mathrm{GHZ}} = \left\langle \mathrm{GHZ}_n \middle| \rho_f \middle| \mathrm{GHZ}_n \right\rangle, \tag{9}$$

where ρ_f is the final density matrix.

Table I compares the energy errors with and without simulated quantum noise, for different evolution times T and numbers of Trotter steps M.

TABLE I. Comparison of VQE and AQC for four qubits with $J = 1$, $h = 0$, and $g = 0.01$. Noise is simulated using the FakeSherbrooke-derived noise model.

	ΔE		F_{GHZ}	
Method (T, M)	Ideal	Noisy	Ideal	Noisy
VQE	7.09×10^{-5}	5.60×10^{-2}	0.4170	0.2900
AQC (20, 100)	4.35×10^{-4}	1.72	0.99988	0.4306
AQC (50, 250)	8.86×10^{-5}	2.99	0.99996	0.1827
AQC (100, 500)	2.58×10^{-5}	3.76	0.99997	0.0706

In the ideal case, VQE gives a small energy error, but a relatively low GHZ fidelity. AQC, on the other hand, achieves a fidelity close to unity, even when its energy error is larger than that of VQE. This illustrates that a small energy error does not necessarily imply a high fidelity, particularly when the lowest energy levels are close together.

With noise, VQE maintains a relatively small energy error, while AQC becomes less accurate as the number of Trotter steps increases.

VQE noise analysis

To investigate the sensitivity of VQE to quantum noise, phase-flip errors are introduced on RY gates and depolarizing errors on CX gates (cf. Fig. 1).

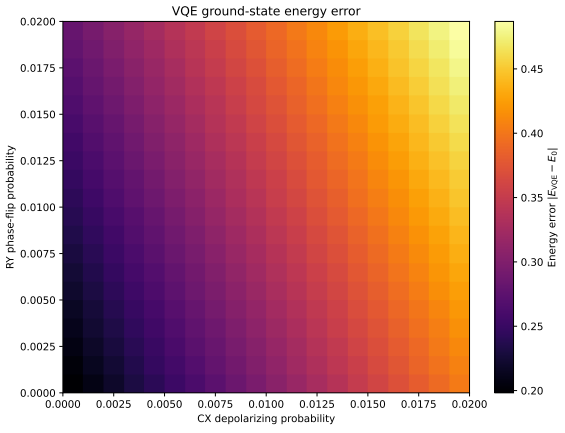


FIG. 2. Absolute VQE ground-state energy error as a function of RY phase-flip and CX depolarizing probabilities.

For the four-qubit Ising model with $J = 1$, $h = 0$, and $g = 1$, the corresponding error probabilities p_{RY} and p_{CX} are varied independently between 0 and 0.02.

At each point, VQE is performed and the resulting ground-state energy is compared with the exact value. The results are shown in Fig. 2.

From Fig. 2, the energy error increases mainly with the CX depolarizing probability, while the RY phase-flip probability has a smaller effect within the considered range.

The observed energy error contains contributions from both quantum noise and imperfect variational optimization. Consequently, a nonzero error is also present in the absence of noise.

Adiabatic quantum evolution

The performance of adiabatic quantum evolution is investigated by varying the total evolution time T and the number of Trotter steps M. The final-state GHZ fidelity is calculated for ideal evolution and for evolution with a noise model derived from the FakeSherbrooke backend. Fig. 3 and Fig. 4 both display the final-state GHZ fidelity with and without simulated backend noise.

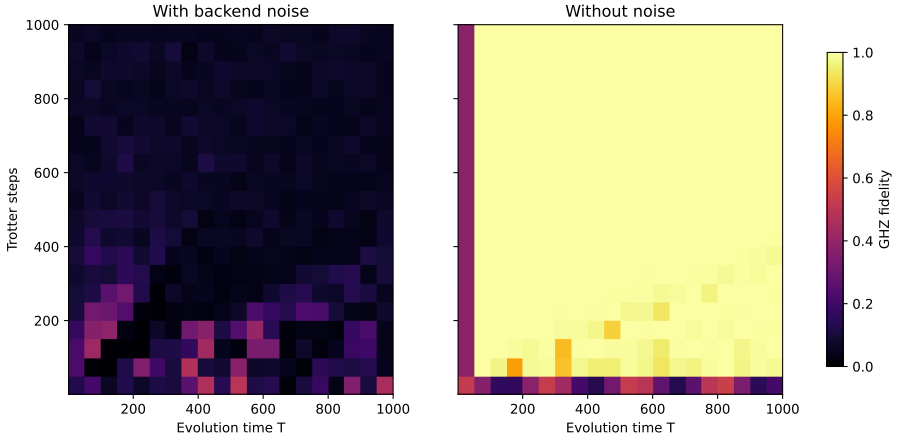


FIG. 3. GHZ fidelity as a function of evolution time T and number of Trotter steps M, with noise (left) and without noise (right). Both panels use the same color scale.

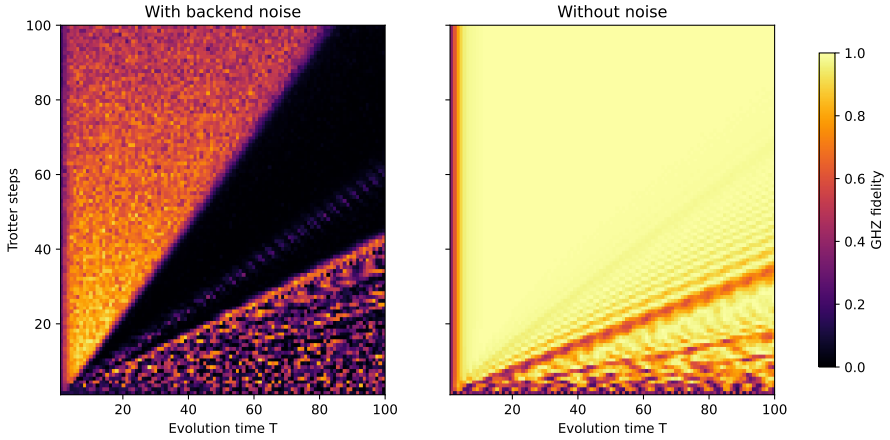


FIG. 4. GHZ fidelity as a function of evolution time T and number of Trotter steps M, with noise (left) and without noise (right). Both panels use the same color scale.

In Fig. 3, $M, T \in [1, 1000]$, using a 20×20 grid. Without noise, the fidelity approaches unity as both M and T increase. With noise, on the other hand, the fidelity decreases significantly for larger M, particularly for $M \gtrsim 200$.

Fig. 4 shows the GHZ fidelity for $M, T \in [1, 100]$, using a 100×100 grid. Without noise, the fidelity approaches unity over much of the parameter space, although regions of lower fidelity appear for small numbers of Trotter steps. This agrees with Fig. 3.

With noise, however, high fidelity is restricted to smaller region, approximately $T \in [5, 32]$ and $M \in [5, 40]$. The highest observed fidelity, $F_{\mathrm{GHZ}} = 0.951$, is achieved with $T = 10$ and $M = 15$.

Increasing M at fixed T improves the Trotter approximation but also increases the number of noisy quantum gates. Conversely, increasing T at fixed M increases the time step $\Delta t = T/M$, potentially introducing larger Trotter errors.

The results therefore illustrate the trade-off between adiabatic evolution, Trotter approximation errors, and accumulated quantum noise.

DISCUSSION AND CONCLUSION

VQE and digitized adiabatic evolution have been compared using the transverse-field Ising model.

In the ideal simulations, both methods give good estimates of the ground-state energy, although AQC achieves a significantly higher GHZ fidelity for the parameters considered. With simulated noise, however, VQE gives smaller energy errors, while AQC is strongly affected by increasing circuit depth.

The results demonstrate the importance of considering quantum noise when comparing quantum algorithms. The simulations are limited to small systems, and further work could include larger systems and methods for reducing the effects of noise.

[1] I. C. O. Rukan, Ising Model: VQE and Adiabatic Quantum Evolution with Noise, GitHub repository (2026), accessed: 20 October 2026.

[2] IBM Quantum, FakeSherbrooke, IBM Quantum documentation, accessed: 20 October 2026.